local-llm-import-export

Quick Start Guide

What Is This?

local-llm-import-export packages your documents and a local language model into a single portable zip file. Import that zip on any computer and chat with your knowledge base completely offline - no internet, no API key, no cloud.

Step 1 - Export (http://localhost:5173)

- Open the export app in Chrome 113+ at http://localhost:5173
- Select a chat model: TinyLlama (700 MB), Gemma 2B (1.4 GB), or Phi-3-mini (2.2 GB)
- Edit the system instructions textarea to define the assistant persona
- Drop your documents (PDF, DOCX, TXT, MD) into the file drop zone
- Click Start Export, choose a save location - streams directly to disk
- Allow 5-20 minutes depending on model size

Step 2 - Import (http://localhost:5174)

- Copy the zip to the target machine
- Open the import app in Chrome 113+ at http://localhost:5174
- Drop the zip onto the page - extraction runs in a background worker
- Wait for GPU upload to complete (2-10 minutes)
- Chat is now fully offline

Supported Models

TinyLlama 1.1B - 700 MB zip, fastest, basic quality
Gemma 2B it - 1.4 GB zip, recommended
Phi-3-mini 4k - 2.2 GB zip, best quality, needs dedicated GPU

Browser Requirements

- Chrome 113+ or Edge 113+ on desktop
- WebGPU available (navigator.gpu)
- Secure context: localhost or https only
- GPU with 1 GB+ VRAM (4 GB+ for larger models)
- Cross-origin isolation enabled automatically by service worker

System Instructions

The System Instructions textarea defines how the chat assistant behaves. A helpful default is pre-filled. Drop an instructions.md file to append content to it. Instructions are stored in manifest.json and used as the LLM system prompt when chatting in the import app.

Troubleshooting

- File picker fails: showSaveFilePicker must be called before any async work
- Export OOM: model shards must never load into main thread RAM

- Import stuck at 0%: check browser console for unzip-worker errors
- GPU upload hangs: try a smaller model or check WebGPU support
- Irrelevant chat answers: check system instructions and source citations
- SharedArrayBuffer missing: hard refresh to activate COI service worker

Verify All APIs in Console

Open DevTools (F12) and run:

```
console.log({ webgpu: !!navigator.gpu,
  filePicker: "showSaveFilePicker" in window,
  coi: crossOriginIsolated,
  sab: typeof SharedArrayBuffer !== "undefined" })
```

All four values should be true.